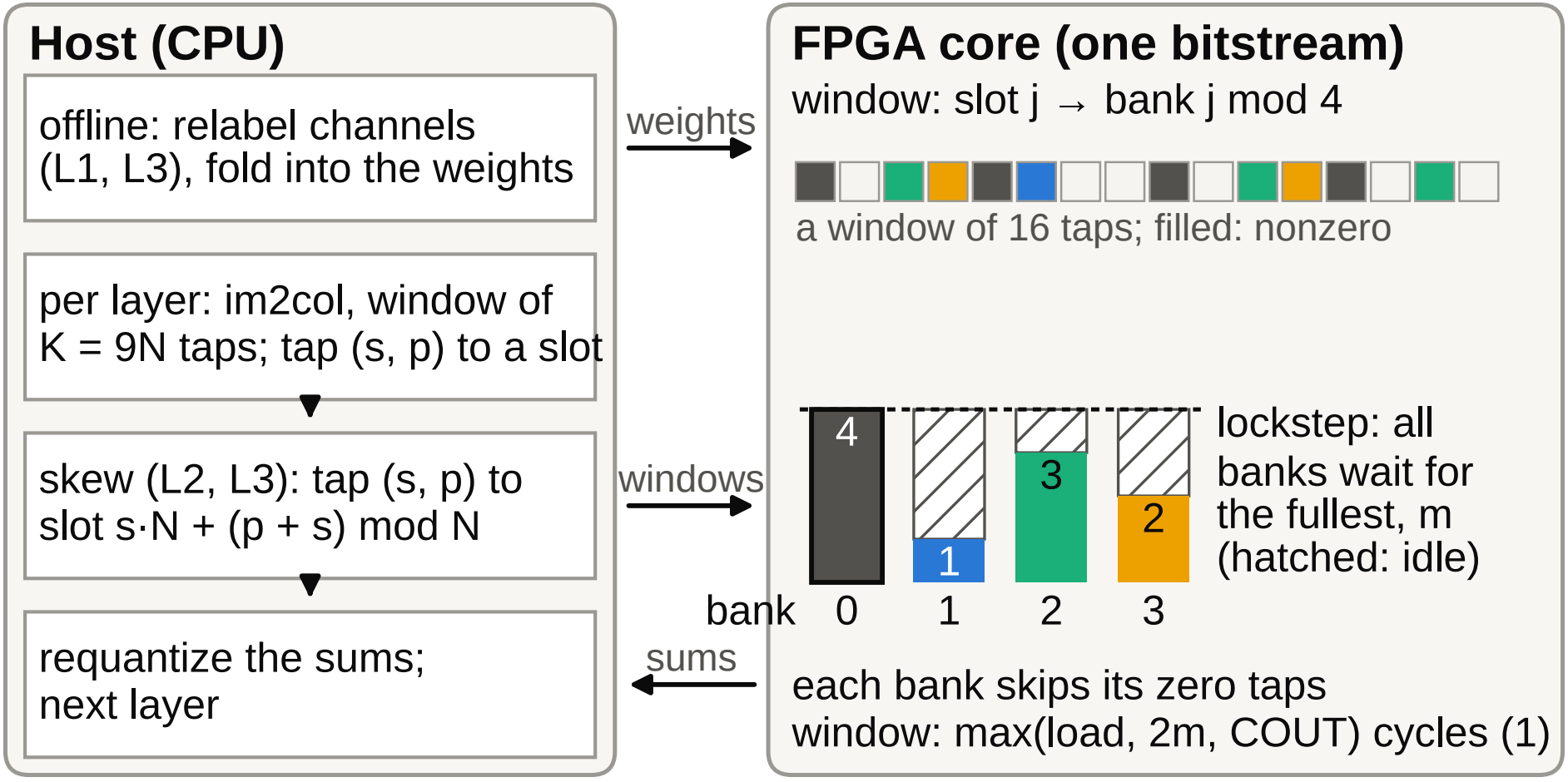
Host (CPU)
offline: relabel channels (L1, L3), fold into the weights
per layer: im2col, window of K = 9N taps; tap (s, p) to a slot
skew (L2, L3): tap (s, p) to slot s·N + (p + s) mod N
requantize the sums; next layer
weights
windows
sums
FPGA core (one bitstream)
window: slot j → bank j mod 4
a window of 16 taps; filled: nonzero
4
1
3
2
lockstep: all banks wait for the fullest, m (hatched: idle)
bank 0 1 2 3
each bank skips its zero taps
window: max(load, 2m, COUT) cycles (1)